

👉 XAI: Post-hoc - SmoothGrad & Integrated Gradients

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**eXplainable
Artificial Intelligence**

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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<https://interpretable-ml-class.github.io/>

SmoothGrad: removing noise by adding noise

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Abstract

Explaining the output of a deep network remains a challenge. In the case of an image classifier, one type of explanation is to identify pixels that strongly influence the final decision. A starting point for this strategy is the gradient of the class score function with respect to the input image. This gradient can be interpreted as a *sensitivity map*, and there are several techniques that elaborate on this basic idea. This paper makes two contributions: it introduces SMOOTHGRAD, a simple method that can help visually sharpen gradient-based sensitivity maps, and it discusses lessons in the visualization of these maps. We publish the code for our experiments and a website with our results.

A common approach to understanding the decisions of image classification systems is to find regions of an image that were particularly influential to the final classification. (Baehrens et al., 2010; Zeiler & Fergus, 2014; Springenberg et al., 2014; Zhou et al., 2016; Selvaraju et al., 2016; Sundararajan et al., 2017; Zintgraf et al., 2016). These approaches (variously called sensitivity maps, saliency maps, or pixel attribution maps; see discussion in Section 2; use occlusion techniques or calculations with gradients to assign an “importance” value to individual pixels which is meant to reflect their influence on the final classification.

In practice these techniques often do seem to highlight regions that can be meaningful to humans, such as the eyes in a face. At the same time, sensitivity maps are often visually noisy, highlighting some pixels that—to a human eye—seem randomly selected. Of course, a priori we cannot determine if this noise reflects an underlying truth about how

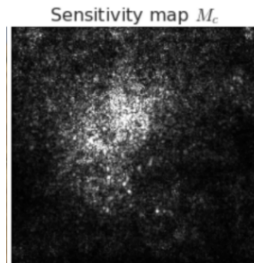
(Smilkov et al. 2017)

🍷 Motivation ---

We want **post-hoc explanations of image classifiers**

Solution:

- **Sensitivity maps** (a.k.a. saliency maps, pixel attribution maps)
- Visual interpretation of gradient of class activation function w.r.t input image
 - ▶ Structured as grayscale image w/ dimension same as input image
 - ▶ Brightness of pixel $\propto$ importance to classification decision



🍷 Background: Gradients as Sensitivity Maps ---

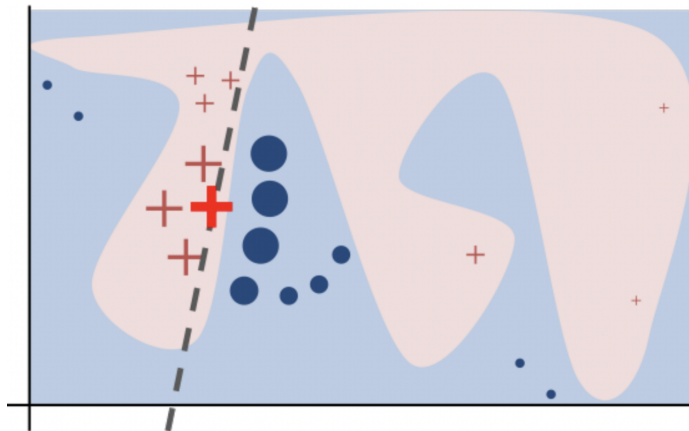
- N : network that classifies images into one class from set C
- Given input image x , N typically computes class activation function S_c for each $c \in C$
- Final classification $\text{class}(x) = \arg\max_{c \in C} S_c(x)$
- **Sensitivity map given by**

$$M_c(x) = \partial S_c(x) / \partial x$$

- **Intuition:** M_c represents how much difference a tiny change in each pixel of x would make to the classification score for class c .
- In practice, this is done with **backpropagation** — same as during training, but instead of updating weights, we stop at the input pixels. The result is a gradient map with one value per pixel.

🍷 Related Work: Perturbation Methods ---

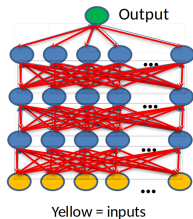
- **Key idea:** generate a perturbed dataset to fit an explainable model
 - ▶ LIME
 - ▶ KernelSHAP



🍷 Related Work: Backpropagation ---

Smoothgrad lives in the family of backpropagation-based methods for explaining neural networks.

All methods propagate a signal backwards from the output to the inputs to assign importance scores to pixels — just like training, but stopping at the input. They differ in **how** they modify that signal along the way:



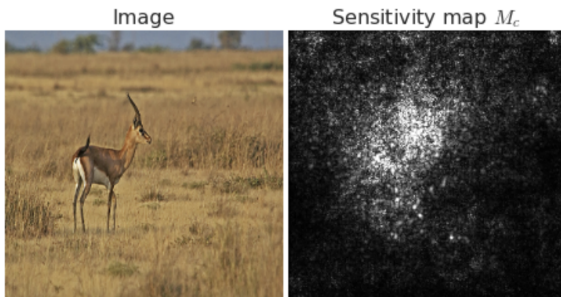
- **Vanilla gradients** — raw gradient $\partial S_c / \partial x$, no modifications. Baseline for SmoothGrad.
- **Integrated Gradients** — averages gradients along a path from a reference image to x .
- **DeepLIFT** — propagates activation *differences* relative to a reference, instead of raw gradients.
- **LRP** — redistributes output relevance backwards layer by layer using conservation rules.
- **Guided Backprop / Deconvolution** — discards negative gradients through ReLUs to highlight only positive contributions.

SmoothGrad is complementary, not a replacement — it can be applied on top of any of these to reduce visual noise.

🍷 Limitations of Sensitivity Maps ---

Visually noisy

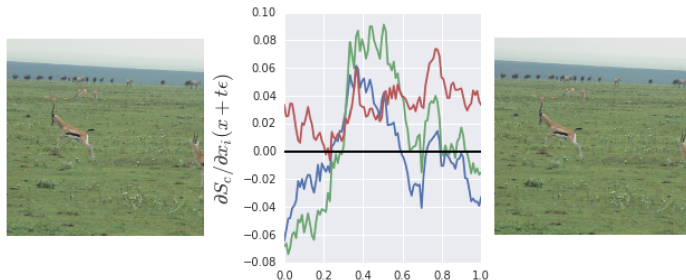
- Often highlight pixels that—to a human eye—seem randomly selected
- *a priori*, we cannot know if this noise reflects an underlying truth about how networks perform classification, or is due to more superficial factors
- This is why saliency maps are typically visualized as a heatmap-like plot.



🍷 Theory Behind SmoothGrad: Noisy Gradients ---

Key idea: Noisy maps are due to noisy gradients

- The derivative of S_c may fluctuate sharply at small scales — networks use **ReLU**s, so S_c is **not even continuously differentiable**.
- *Two images that look identical to a human can have very different gradients — so the raw gradient at a single point is not a reliable importance signal.*
- The noise you see in the map is not necessarily telling you something meaningful it may just be an accident of the mathematical landscape of S_c



SmoothGrad: Intuition ---

Simple solution

- take an image of interest
- sample similar images by adding Gaussian noise to the image
- take the average of the resulting sensitivity maps for each sampled image
- This smoothes the gradient

SmoothGrad: Algorithm ---

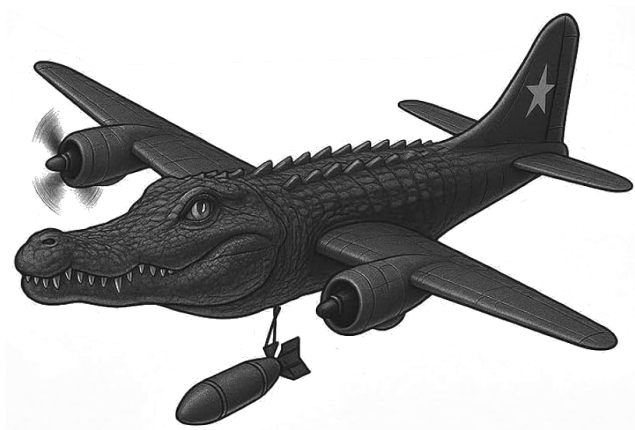
- 1 Take random samples in a neighborhood of an input x with added noise
- 2 Average the resulting sensitivity maps

$$\hat{M}_c(x) = \frac{1}{n} \sum_1^n M_c(x + \mathcal{N}(0, \sigma^2))$$

- n is the number of samples,
- $\mathcal{N}(0, \sigma^2)$ represents Gaussian noise with standard deviation σ .

🍷 Experimental Setup ---

- Performed SmoothGrad on visualizations of two neural networks:
 - ▶ Inception v3 model by Google that was trained on the ILSVRC-2013 dataset
 - ▶ Convolutional MNIST model based on the TensorFlow tutorial



🍷 Choosing Hyperparameters (σ : std. dev.) ---

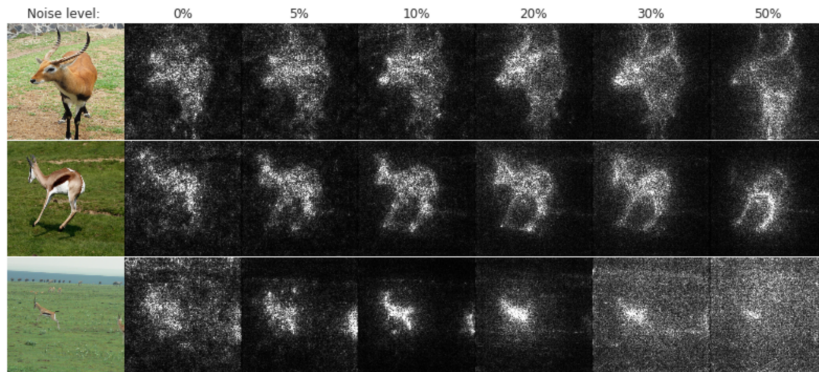


Figure 3. Effect of noise level (columns) on our method for 5 images of the gazelle class in ImageNet (rows). Each sensitivity map is obtained by applying Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the input pixels for 50 samples, and averaging them. The noise level corresponds to $\sigma/(x_{max} - x_{min})$.

$$\hat{M}_c(x) = \frac{1}{n} \sum_1^n M_c(x + \mathcal{N}(0, \sigma^2))$$

σ : the standard deviation of the Gaussian noise

🍷 Choosing Hyperparameters (n : sample size) ---

$$\hat{M}_c(x) = \frac{1}{n} \sum_1^n M_c(x + \mathcal{N}(0, \sigma^2))$$

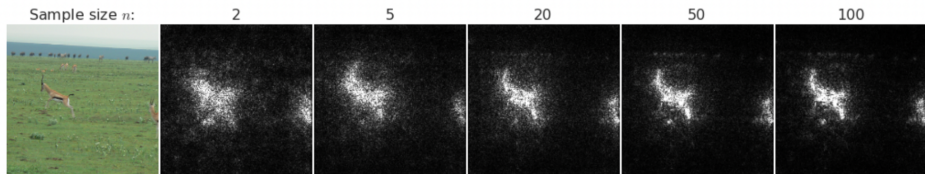


Figure 4. Effect of sample size on the estimated gradient for inception. 10% noise was applied to each image.

🍷 Qualitative Results: Visualization Techniques ---

● Absolute Value of Gradients

- ▶ depends on the characteristics of dataset



● Capping outlying values

- ▶ presence of few pixels that have much higher gradients than the average
- ▶ capping to 99 percentile

● Multiplying maps with images

- ▶ produce simpler & sharper images (Shrikumar et al., 2017; Sundararajan et al., 2017)
- ▶ Downside: Pixels with values of 0 will never show up on the sensitivity map.
- ▶ Upside: when viewing the importance of the feature as contribution to the image

🍷 Qualitative Results: Visual Coherence 1/2 ---

Definition

Visual Coherence Highlights are only on the object of interest, not the background

Comparison with three gradient-based methods

- Vanilla gradient
- Integrated Gradients
- Guided BackProp



Qualitative Results: Visual Coherence 2/2 ---

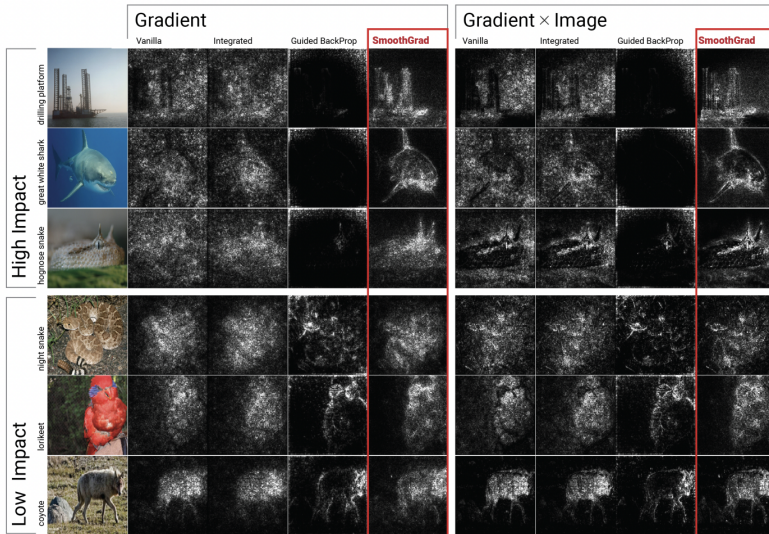


Figure 5. Qualitative evaluation of different methods. First three (last three) rows show examples where applying SMOOTHGRAD had high (low) impact on the quality of sensitivity map.

🍷 Qualitative Results: Discriminativity ---

Definition

Discriminativity the ability to explain / distinguish separate objects without confusion

Open Problem

Which properties affect the discriminativity of a given methods?

- Why did GBP show the worst performance?

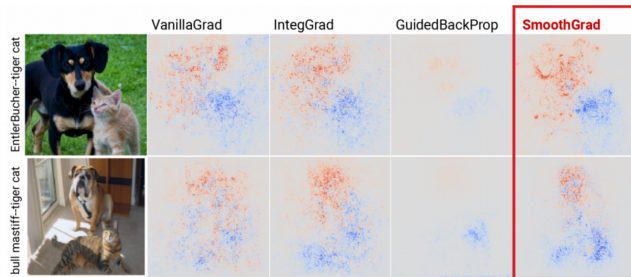
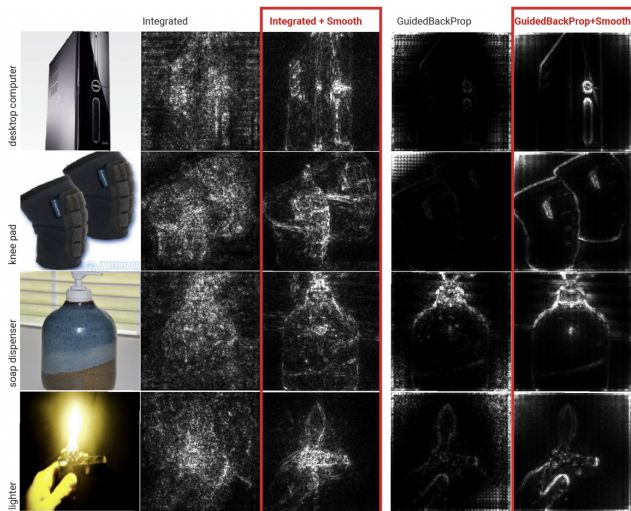


Figure 6. Discriminativity of different methods. For each image, we visualize the difference $\text{scale}(\partial y_1 / \partial x) - \text{scale}(\partial y_2 / \partial x)$ where y_1 and y_2 are the logits for the first and the second class (i.e., cat or dog) and $\text{scale}()$ normalizes the gradient values to be between $[0, 1]$. The values are plotted using a diverging color map $[-1, 0, 1] \mapsto [\text{blue}, \text{gray}, \text{red}]$. Each method is represented in columns.

🍷 Combining with Other Methods ---

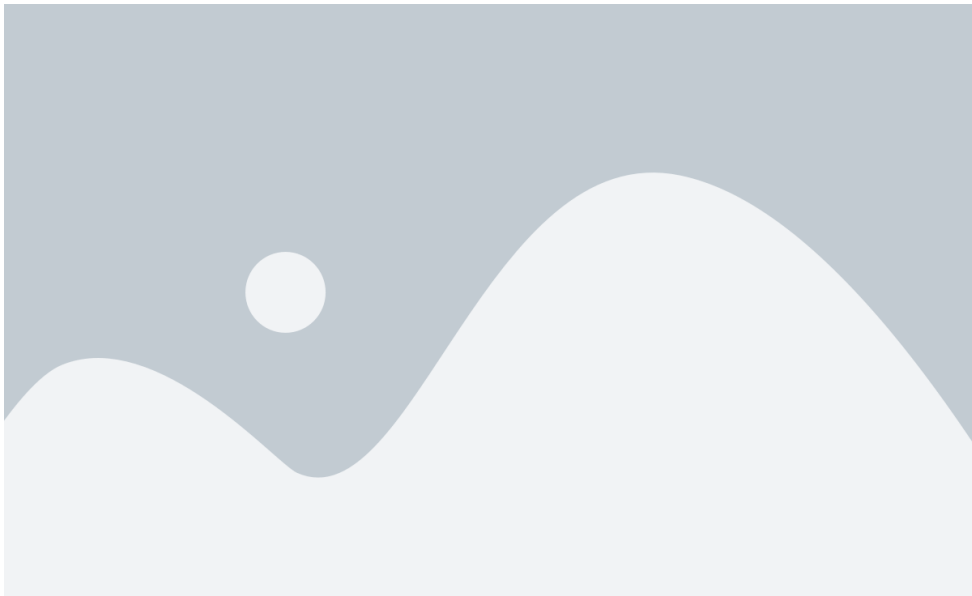
The same smoothing procedure can be used to augment any gradient-based method.



🍷 Limitations / Discussion ---

- Completely qualitative results, can we get **quantitative metrics**?
- Noisy sensitivity maps are **due to noisy gradients**
 - ▶ Is this true?
 - ▶ Future work: look for further evidence and theoretical arguments
- Does SmoothGrad **generalize** to other networks & tasks?
- How do we tradeoff between making picture pretty and being faithful to the model? Do you think SmoothGrad handled this tradeoff well?

🍷 Paper 2 ---



🍷 Motivation and Problem Statement ---

- Feature Attribution:

Definition

Formally, suppose we have a function $F : \mathbb{R}^n \rightarrow [0, 1]$ that represents a deep network, and an input $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. An attribution of the prediction at input x relative to a baseline input x' is a vector $A_F(x, x') = (a_1, \dots, a_n) \in \mathbb{R}^n$ where a_i is the contribution of x_i to the prediction $F(x)$.



- Examples: in a CNN an attribution method could reveal which pixels were responsible for a certain label being picked (we saw this with LIME/SHAP)
- Problem: attribution technique are hard to evaluate empirically - hard to separate errors from model vs errors from attribution method
 - ▶ Ex. Gradients
 - ▶ Baseline: black image, empty text, etc.

Summary of Contributions ---

- Present two axioms: **Sensitivity** and **Implementation Invariance**
 - ▶ **Sensitivity:** For every input and baseline that differ in one feature but have different predictions then the differing feature should be given a non-zero attribution.
 - ▶ **Implementation Invariance:** The attributions are always identical for two functionally equivalent networks.
- 2 axioms $\rightarrow$ **integrated gradients**
 - ▶ Overview: path integral of the gradients along the straight line path from an input x to a baseline input x'

🍲 Two Axioms (Desiderata) ---

Sensitivity (a)

Definition: When 2 inputs that differ in only one feature result in different predictions, the **differing feature** should be given a **non-zero attribution**.

Invariance

Definition: The attributions are always identical for two functionally equivalent networks.

$$\frac{\partial f}{\partial g} = \frac{\partial f}{\partial h} \cdot \frac{\partial h}{\partial g}$$

🍷 Other Attribution Methods ---

Gradients (of the output with respect to the input)

- Breaks sensitivity - prediction function can flatten at the input, giving 0 gradient despite function value at the input being different from the baseline
- Example:
 - ▶ Single ReLU network: $f(x) = 1 - \text{ReLU}(1 - x)$
 - ★ Baseline: $x = 0$, input: $x = 2$
 - ★ $f(0) = 0$, $f(2) = 1$
 - ★ Since f is flat at $x = 1$, gradient gives attribution of 0 to x

🍷 Other Attribution Methods: Methods that Break Sensitivity ---

Methods that Break Sensitivity

- DeConvNets, Guided back-propagation



🍷 Other Attribution Methods: Methods that Break Implementation Invariance ---

Methods that Break Implementation Invariance

- DeepLift and Layer-wise relevance propagation (LRP)



The Method

Integrated Gradients ---

Definition

The **path integral** of the gradients along the **straight-line path** from the baseline x' to the input x .

$$\text{IntegratedGrads}_i(x) := (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha$$

New Axiom

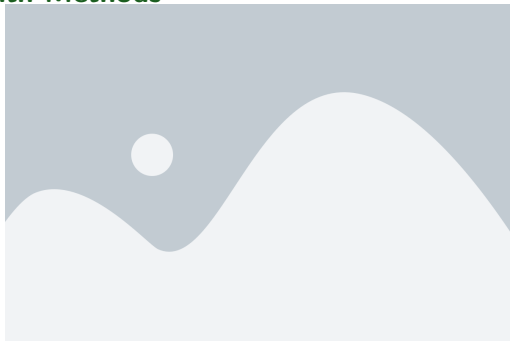
Completeness: The sum of the attributions is equal to the difference of the outputs.

Proposition 1. If $F : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable almost everywhere¹ then

$$\sum_{i=1}^n \text{IntegratedGrads}_i(x) = F(x) - F(x')$$

🍷 Uniqueness of Integrated Gradients ---

Path Methods



$\text{PathIntegratedGrads}_i^\gamma(x) :=$

$$\int_{\alpha=0}^1 \frac{\partial F(\gamma(\alpha))}{\partial \gamma_i(\alpha)} \frac{\partial \gamma_i(\alpha)}{\partial \alpha} d\alpha$$

Axioms

- **Sensitivity (b):** If the function does not depend (mathematically) on some input, then the attribution for that input is always zero. -
- **Linearity:** Attributions preserve any linearity within the network.

$$a \times f_1 + b \times f_2$$

- **Symmetry-Preserving:** For symmetric variables, if they have identical values in the input and identical values in the baseline, they then receive identical attributions.

$$\text{Si } F(x, y) = F(y, x)$$

🍷 Using Integrated Gradients ---

Selecting a Baseline

****Two Components:****

- Zero-Score: $F(x') \approx 0$ - Conveys Absence of Signal

****Examples:****

- Object Recognition: All-black image - Text: All-zero input embedding vector

Computing IGs

$\text{IntegratedGrads}_i^{\text{approx}}(x) :=$

$$(x_i - x'_i) \times \sum_{k=1}^m \frac{\partial F(x' + \frac{k}{m} \times (x - x'))}{\partial x_i} \times \frac{1}{m}$$

Experimental Results

🍷 Object Recognition CNN ---

Task: Given image, predict the category of the object



🍷 Question Classification CNN ---

Task: Given question, predict what type of answer it is looking for.



🍷 Machine Translation RNN ---

Task: Given English sentence, predict German translation



🍷 Ligand Screening Graph CNN ---

Task: Given molecular graph, predict whether it is active against an enzyme



🍷 Conclusion and Discussion ---

Summary

- Formalizes two axioms for attribution: **sensitivity, implementation invariance**
- Propose **integrated gradients** and argue that it is theoretically superior to other gradient-based methods (e.g. DeepLift, LRP, guided backprop, etc.)
- Perform experiments across several domains to showcase method

Discussion Questions

- Are you convinced that these axioms are desirable?
- Do you see any strengths or weaknesses in the idea of producing explanations through an integrated path?
- Have the experiments convinced you of the superiority of their method?

Thank You!

References --- |

- Smilkov, Daniel, Nikhil Thorat, Been Kim, Fernanda Viégas, and Martin Wattenberg. 2017. “SmoothGrad: Removing Noise by Adding Noise.” *arXiv Preprint arXiv:1706.03825*.
- Sundararajan, Mukund, Ankur Taly, and Qiqi Yan. 2017. “Axiomatic Attribution for Deep Networks.” In *International Conference on Machine Learning*, 3319–28. PMLR.